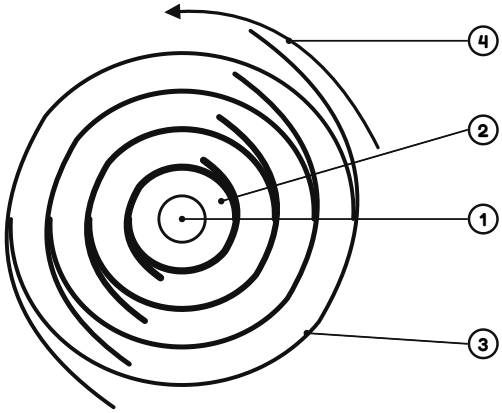
EARTH AND SPACE

Storms, Cyclones and Climate Change

How tropical storms and cyclones form, their effects, tracking and warnings, and how people are changing the climate



In this pack

- **Learn it:** Storms, cyclones and our changing climate ★☆☆
- **Fact check:** Cyclone facts ★☆☆
- **Think it through:** Cyclones and climate ★★☆
- **Investigate:** Track a cyclone ★★☆
- **Real-world case study:** Ready for the storm: a family cyclone plan ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Storms, cyclones and our changing climate

Name: Class: Date:

Learning goals

- Explain how tropical depressions, storms and cyclones form.
- Describe how cyclones affect people, animals and plants, and how warnings keep people safe.
- Explain how human actions are changing weather patterns and the climate.

Bangladesh sits at the top of the **Bay of Bengal**, a warm sea shaped like a funnel. Several times a year, huge spinning storms called **cyclones** form over this warm water. Some of them travel north towards our coast. Learning how they form, and tracking where they go, saves thousands of lives.

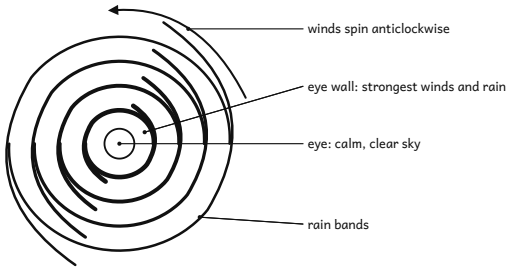
How a cyclone is born

- The Sun heats the sea. When the water is warmer than about **27 °C**, lots of water **evaporates**.
- Warm, damp air **rises** quickly. As it rises it cools and the water vapour **condenses** into huge clouds.
- More air rushes in to take its place, and because the Earth is spinning, the air begins to **spin** too.
- If the sea stays warm, the storm keeps growing: from a **low pressure area** to a **depression**, a **cyclonic storm**, and finally a **severe cyclone**.

Name	Wind speed (about)
depression	up to 61 km/h
cyclonic storm	62 – 88 km/h
severe cyclonic storm	89 – 117 km/h
very severe cyclone	118 km/h or more

Around the world

The same kind of storm has different names: a **cyclone** in the Bay of Bengal and Indian Ocean, a **hurricane** in the Atlantic, and a **typhoon** in the Pacific.



A cyclone seen from above.

Why are cyclones so dangerous?

- **Strong winds** blow down houses, trees and power lines.
- **Heavy rain** causes floods and landslides.
- A **storm surge**, a wall of sea water pushed onto the land by the wind, floods villages and makes the soil salty.

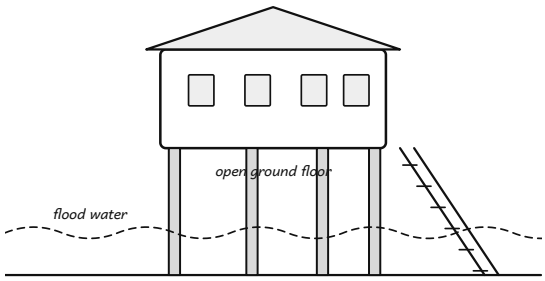
Who is affected	How
people	homes and crops destroyed, injuries, no clean water or electricity
animals	cattle drown, wild animals lose their homes, fish ponds flood with salt water
plants	trees uprooted, crops flattened, salty water kills rice and vegetables

Tracking a cyclone

Weather satellites take pictures of the clouds every few minutes. Scientists mark the centre of the cyclone on a map using **latitude** (how far north) and **longitude** (how far east). Joining the points shows the cyclone's **track** and helps predict where it will reach land.

Warning signals

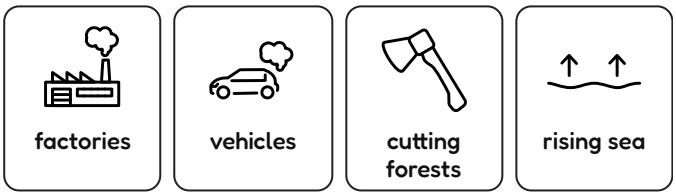
Bangladesh uses **warning signals numbered 1 to 11** for the sea ports. The higher the number, the greater the danger. Signals 8, 9 and 10 mean **great danger**: people in the coastal areas must go to a **cyclone shelter**. Volunteers spread the warning with megaphones, flags and mobile phone messages.



A cyclone shelter stands on tall pillars above the storm surge.

Our changing climate

Burning **coal, oil and gas** in cars, factories and power stations, and **cutting down forests**, add more **carbon dioxide** to the air. It acts like a blanket and traps heat, so the Earth is slowly getting **warmer**. Warmer seas can make cyclones **stronger** and wetter, melting ice makes the **sea level rise**, and rainfall becomes harder to predict.



Bangladesh connection

Bangladesh produces only a tiny share of the world's carbon dioxide, but it is one of the countries most at risk from climate change, because so much of the land is low and close to the sea.

Key words

cyclone	a huge spinning storm that forms over warm sea
depression	an area of low pressure; a weak tropical storm
storm surge	sea water pushed onto land by a cyclone
track	the path a cyclone follows
latitude	how far north or south a place is
longitude	how far east or west a place is
cyclone shelter	a strong building on pillars where people stay safe
climate change	long-term change in the Earth's weather patterns

Cyclone facts

★☆☆
Grade 4

Name: Class: Date:

Learning goals

- Order the stages of a cyclone.
- Label a cyclone.
- Recall safety facts.

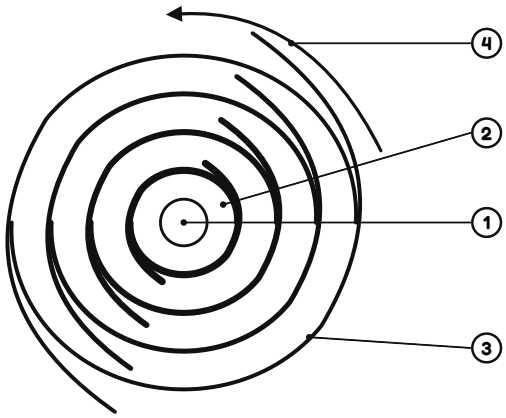
1 Number the steps in the order a cyclone forms. [4]

- ☐ The storm grows into a cyclone.
- ☐ The Sun heats the sea above 27 °C.
- ☐ Air rushes in and starts to spin.
- ☐ Water evaporates and warm, damp air rises.
- ☐ Huge clouds form as the vapour condenses.

2 Label the cyclone. [4]

LABELS

rain bands **eye: calm, clear sky** **winds spin anticlockwise**
eye wall: strongest winds and rain



1 _____ 2 _____

3 _____ 4 _____

3 Match the danger to what it does. [3]

Draw a line to match each pair, or write the letter.

strong wind ●	● **A** causes floods and landslides
heavy rain ●	● **B** pushes salty sea water onto the land
storm surge ●	● **C** blows down trees and houses

4 True or false? [4]

Statement	True	False	If false, correct it
1. Cyclones form over cold water.			
2. Signal 10 means great danger.			
3. A cyclone shelter is built on tall pillars.			
4. Burning fuels adds carbon dioxide to the air.			

Cyclones and climate

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Explain cyclone formation.
- Link human actions to climate change.

1 **Why do most cyclones in the Bay of Bengal happen in April–May and October–November, and not in January?** [2]

2 **Why do cyclones get weaker after they move over land?** [2]

3 **Explain how salty water from a storm surge can cause problems for farmers for many months.** [2]

4 **Complete the chain to show how human actions can make cyclones more dangerous.** [3]

- ☐ cyclones can become stronger
- ☐ people burn coal, oil and gas
- ☐ the sea gets warmer
- ☐ more carbon dioxide in the air
- ☐ heat is trapped and the Earth warms

5 **Suggest two ways people in Bangladesh can adapt to stronger cyclones.** [2]

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INVESTIGATE · B2 · CYCLONES AND CLIMATE

Track a cyclone

★★☆
Grade 4

Name: Class: Date:

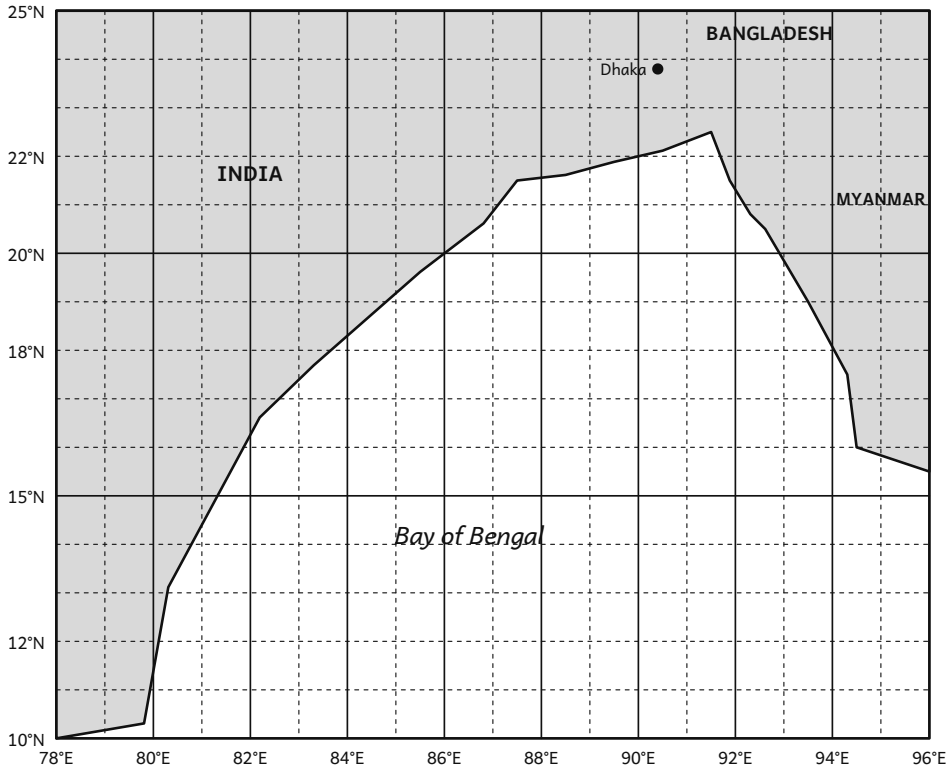
Learning goals

- Plot points using latitude and longitude.
- Describe a track.
- Use data to make predictions.

These are the positions of the centre of Cyclone **Sidr** in November 2007 (rounded). Plot each point on the map, join them in order, and write the date next to each point.

Date	11 Nov	12 Nov	13 Nov	14 Nov	15 Nov	16 Nov
Latitude (°N)	11.5	13	15	17.5	21	24
Longitude (°E)	92	90	89.5	89.3	89.6	90.8

1 **Plot the track.** [4]



2 **In which direction did Sidr mostly travel?** [1]

3 **On which date did it reach the coast of Bangladesh?** [1]

4 **About how far north did it move each day between 13 and 15 November?** [1]

5 **Why is it important to track a cyclone several days before it reaches land?** [2]

REAL-WORLD CASE STUDY · B2 · CYCLONES AND CLIMATE

Ready for the storm: a family cyclone plan

★★★ Grade 4

Name: Class: Date:

Learning goals

- Use real warnings.
- Plan to keep people and animals safe.
- Think about helping others.

Science · Citizenship · Health · Maths

Rahima lives with her family in a village near the coast in Khulna. One evening the radio says the signal has been raised from **4** to **7**, and a cyclone may reach the coast in two days. Volunteers from the **Cyclone Preparedness Programme** walk through the village with megaphones and raise warning flags.

Rahima's family must decide what to take to the shelter, what to do with their goats and chickens, and how to help their grandmother, who cannot walk far. Thanks to warnings, shelters and volunteers, the number of people who lose their lives in cyclones in Bangladesh has fallen enormously since 1970.

Based on the Cyclone Preparedness Programme (Bangladesh Red Crescent Society and the Government of Bangladesh).

1 **Sort the items: which should Rahima's family take to the shelter?** [6]

SORT THESE

dry food **drinking water** **important papers in a plastic bag** **heavy furniture** **medicines** **television**

Take	Leave or protect at home

2 **The signal goes up to 9. What does that mean, and what should the family do?** [2]

3 **How could the family keep their goats and chickens safe?** [1]

4 **How can the family help Grandmother? Who else in the village might need help?** [2]

.....

.....

5 **Design a "cyclone ready" checklist poster for families.** [2]

Cyclone ready!

Show what you know

★★★
Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

O	J	O	D	C	T	K	F	S	Y
S	H	E	L	T	E	R	S	I	K
U	C	C	P	M	Y	J	U	G	W
C	Y	C	L	O	N	E	R	N	I
E	M	Y	I	U	T	W	G	A	N
U	G	G	N	B	R	R	E	L	D
P	F	I	P	E	J	T	E	U	K
N	S	T	O	R	M	N	P	E	J
B	W	A	R	M	K	K	F	Y	H
S	T	T	R	A	C	K	D	E	W

cyclone eye shelter signal storm surge track warm wind

2 Where do cyclones form? [1]

- ◯ over cold mountains
- ◯ over warm seas
- ◯ in deserts
- ◯ over ice

3 The calm centre of a cyclone is called the... [1]

- ◯ eye
- ◯ tail
- ◯ surge
- ◯ track

4 Draw a simple diagram to show how warm, damp air rising from the sea forms storm clouds. [3]

Storm clouds forming

5 **Challenge. Bangladesh is planting mangrove forests along the coast. Explain how this can protect people from cyclones and how it helps the climate.** [3]

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain how a cyclone forms	○	○	○
describe the dangers of wind, rain and storm surge	○	○	○
plot a cyclone track on a map	○	○	○
explain the warning signals	○	○	○
plan how a family can stay safe	○	○	○
explain how human actions change the climate	○	○	○